

Performing 'Yinzer' Dialect in Social Text: An Exploration of /r/Pittsburgh



FOG
WRANGLERS

Will Martin | Ian Williams | Kuheli Sai | Hanh Phan

SICSS CMU 2025 | 05.23.2025



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Will

Transition Design

Data scraping
Data wrangling
Prototyping
Code assembly



Ian

Social Welfare

Literature review
Yinzer lexicon
Slides
Logo



Kuheli

Data Science for Societal Good

Sentiment analysis
Descriptive statistics
Data wrangling

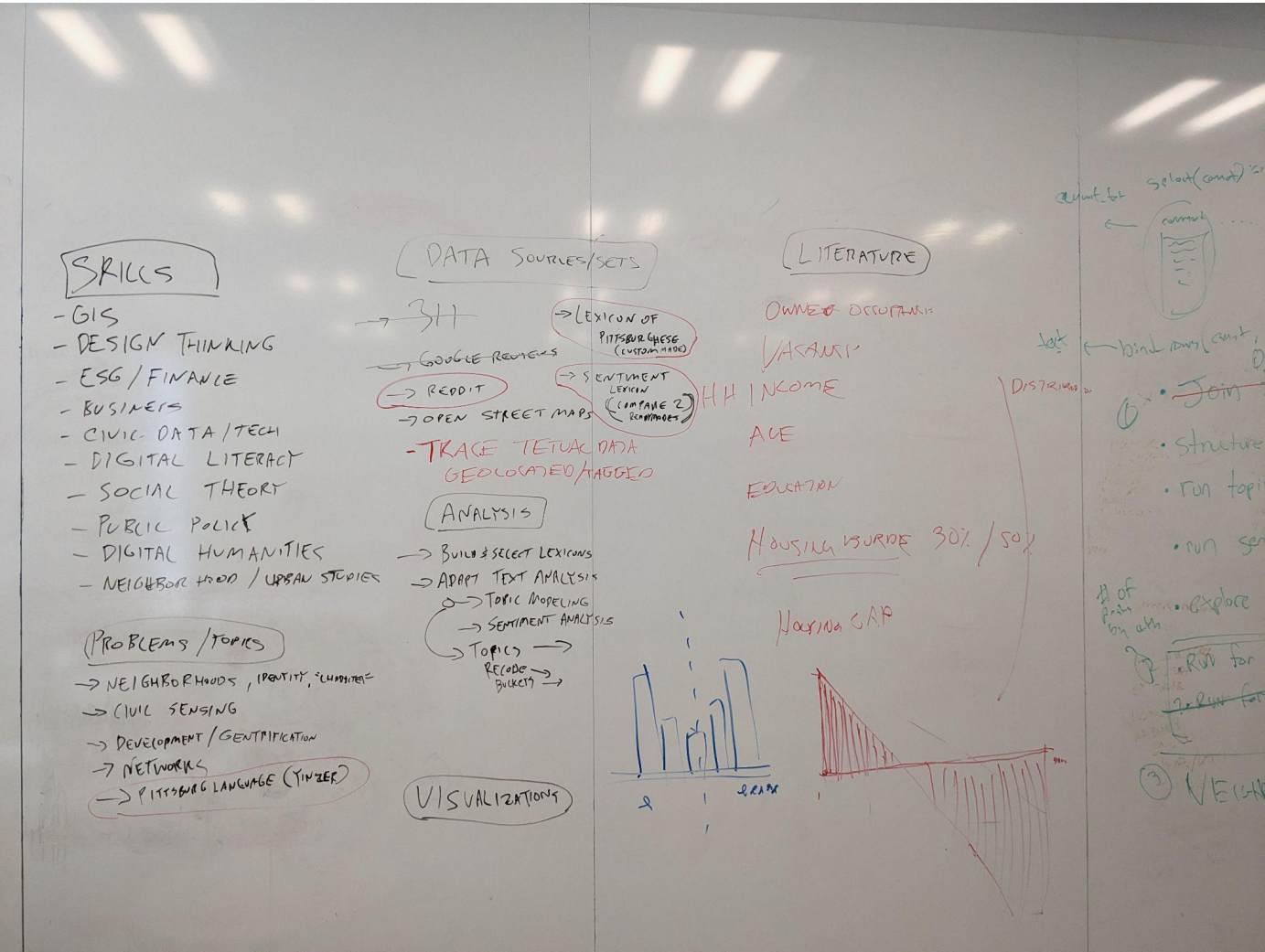


Hanh

Business and Accounting

Topic modeling

Studio



Data Source



/r/Pittsburgh: Pittsburgh's front page of the internet

 Created Apr 7, 2008

 Public

260K
Members

264
 Online

Top 1%
Rank by size 

Corpus:

/r/Pittsburgh threads and comments from
5/4/25-5/20/2025 (two week window)

n = 8177 (266 threads; 7,911 comments)

Largest thread is 497 comments, about
Pittsburgh

Caveat:

Entry limit for Reddit API access

Methods:

Data scraping, text analysis, sentiment analysis,
topic modeling



← → ↺ www.reddit.com/r/pittsburgh/ h. ☆ Search

reddit

Search in r/pittsburgh

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RECENT

r/pittsburgh

TOPICS

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About Reddit

 **r/pittsburgh**

Best Community highlights

What's going on this week? Events/Discussion/Classifieds (Mon-Sun)
10 votes • 8 comments



 **u/Great-Cow7256** • 4 hr. ago

Duquesne Light, West Penn Power officials grilled at hearing over days-long power outages

<https://triblive.com/local/valley-news-dispatch/duquesne-light-west-penn-power-officials-grilled-at...>

110 88 Share

 **u/Quirkykiwi** • 1 mo. ago

My beloved partner's obituary, who was raised in West Mifflin. I came home in Thursday to find him deceased, he was 35 and it was very unexpected. He was a light to all who knew him. Hope it is ok that I share. Ron loved Pittsburgh.

<https://www.legacy.com/us/obituaries/legacyremembers/ronald-murphy-obituary...>



3.1K 248 3 Share

/r/Pittsburgh: Pittsburgh's front page of the internet
Created Apr 7, 2008
Public

260K Members
175 Online
Top 1% Rank by size

COMMUNITY BOOKMARKS

FAQ
City Pictures
Pennsylvania
Looking for friends?
Job searching

RULES

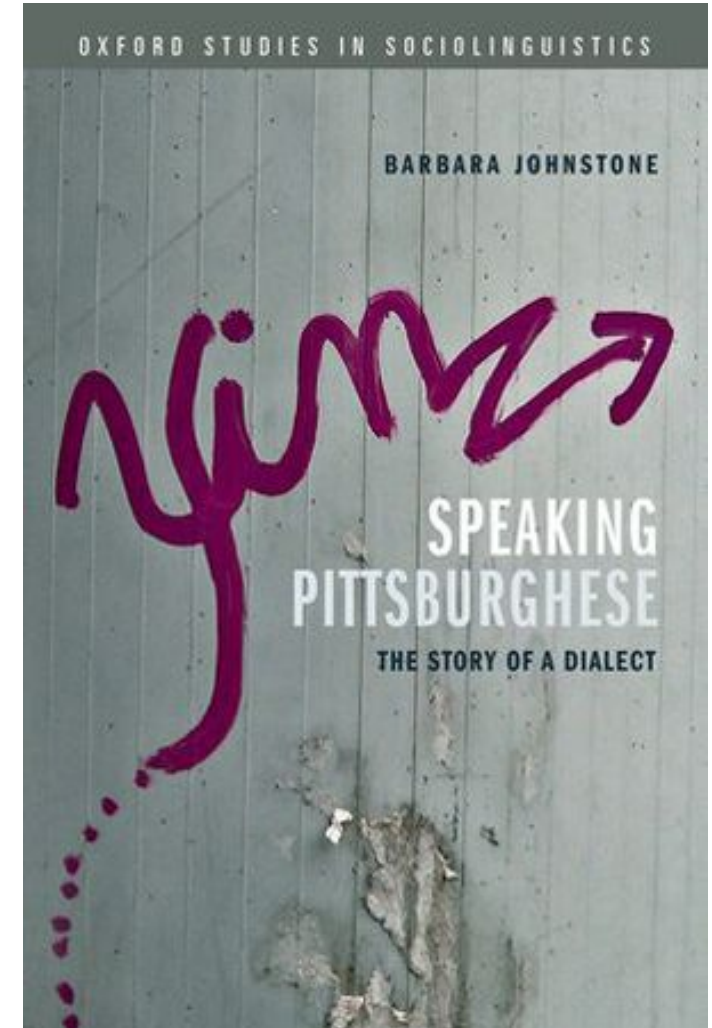
- 1 Location
- 2 Civility
- 3 Hate Speech, Bigotry, and Demeaning Language

pgh_reddit_corpus_2025-0504_2025-0520.csv													
Open with Google Sheets													
	A	B	C	D	E	F	G	H	I	J	K	L	M
1	url	author	date	timestamp	score	upvotes	downvotes	golds	comment	comment_id	thread_author	thead_date	thread_timestamp
2	https://www.reddit.co	NotManny_STG	2025-05-06	1746490332	17	17	0	0	If you received your	1	vjgirl	2025-05-06	174648979
3	https://www.reddit.co	B0bb3r7	2025-05-06	1746490665	12	12	0	0	Let me just slide this **Check if you qualif	2	vjgirl	2025-05-06	174648979
4	https://www.reddit.co	probably_art	2025-05-06	1746493621	8	8	0	0	>You may be able Just to echo that enc I'm pretty sure your r (Pitt is a school, PGH	1	bedhead-deadhead	2025-05-06	174648971
5	https://www.reddit.co	sagittacancers	2025-05-06	1746493225	6	6	0	0	I've not been in eith	2	bedhead-deadhead	2025-05-06	174648971
6	https://www.reddit.co	MammothSuspect20	2025-05-06	1746499668	6	6	0	0	Had a buddy who liv	2	bedhead-deadhead	2025-05-06	174648971
7	https://www.reddit.co	sagittacancers	2025-05-06	1746499768	2	2	0	0	I'd disagree with it be	2_1	bedhead-deadhead	2025-05-06	174648971
8	https://www.reddit.co	MammothSuspect20	2025-05-06	1746500324	5	5	0	0	Do you not have to b	2_1_1	bedhead-deadhead	2025-05-06	174648971
9	https://www.reddit.co	deskcoupon	2025-05-06	1746541023	2	2	0	0	You could bike up B.	2_1_1_1	bedhead-deadhead	2025-05-06	174648971
10	https://www.reddit.co	MammothSuspect20	2025-05-07	1746580881	1	1	0	0	The 75 schedule var	2_2	bedhead-deadhead	2025-05-06	174648971
11	https://www.reddit.co	DPEP56	2025-05-06	1746554144	2	2	0	0	Why not go up bound	2_2_1	bedhead-deadhead	2025-05-06	174648971
12	https://www.reddit.co								I lived across the stre	3	bedhead-deadhead	2025-05-06	174648971
	https://www.reddit.co								Bars for live music: s	4	bedhead-deadhead	2025-05-06	174648971



Framing

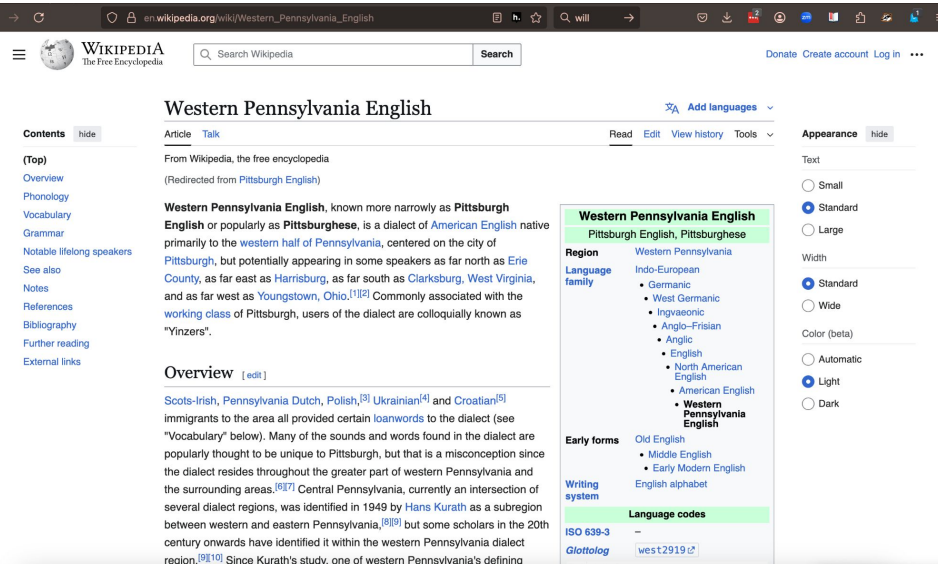
- Reddit and other online communities as sites of place-making, (hyper)local identity, and regional dialects (Berragan, 2024; Eisenstein, 2017; Hamre, 2024; Turner, 2024).
- Reddit as data source and object of scholarly inquiry; user behavior shaped by algorithmic structure of platform, including 'gaming' ratings; researcher behavior shaped by disciplinary norms and assumptions (Proferes et al, 2021).
- Emotion and affect in sentiment analysis not reducible to statistical calculations (Stark, 2019; Umbach and Marshall, 2023).
- Language patterns and lexical triggers as code-switching (Winter et al, 2023).
- Witness as default in many digital platforms (Brunsma et al, 2020).
- Whiteness as default of the 'Yinzer' as paradigmatic Pittsburgh character; emergence of Pittsburghese as local identity started in 1960's (Eberhart 2009; Eberhart, 2012; Johnstone, 2013).



downtown vs dahntahn



In Search of Digital Yinzers



Speakers of Pittsburgh English are sometimes called "Yinzers" in reference to their use of the second-person plural **proun** "yinz." The word "yincer" is sometimes heard as pejorative, indicating a lack of sophistication, but the term is now used in a variety of ways.^[13] Older men are more likely to use the accent than women "possibly because of a stronger interest in displaying local identity...."^[14]

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"Pittsburghese" Overview

What is "Pittsburghese"?

You may be wondering why you haven't seen the term "Pittsburghese" on this website. That is because this website is about Pittsburgh speech—how Pittsburghers actually talk. Pittsburghese, on the other hand, is a reflection of how people THINK Pittsburghers talk.

If you live in the Pittsburgh area, you are surrounded by examples of Pittsburghese, on websites, t-shirts, coffee mugs, souvenir shot glasses, and the like, and in newspaper cartoons and humorous stories about Pittsburghers. There are even dolls that say things in Pittsburghese when you squeeze their stomachs. But these examples don't all reflect things that Pittsburghers actually do in their speech, and many of them are things that people do all over the English-speaking world.

What's the difference between "Pittsburghese" and Pittsburgh speech?

Here is a chart that compares Pittsburgh speech with Pittsburghese.

Feature of Pittsburgh speech, in descriptions by linguists	Item in Sam McCool's New Pittsburghese: How to Speak like a Pittsburgher
Pronunciation	
The vowel in words like "cot" and "caught" is pronounced the same way.	"Dawn" for "Don"

A book cover for "Pittsburghese: How to Speak Like a Pittsburgher" by Sam McCool. The cover features a black and white photo of the Pittsburgh skyline with speech bubbles containing words like "WORSH", "DAHN TAHN", "IMP in ARN", and "YUNZ".

https://pittsburghspeech.pitt.edu/PittsburghSpeech_PgheseOverview.html

Keyword
ahio
Ahz
Arn
ats
babushka
boilermaker
bumbands
carbon oil
carline
cawd
chipped ham
chitchat
cittent
city chicken
couttent
crewsants
cupboard
dahntahn
Dawn
et
felled
filum
grommed
how's about
in regards to

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 r/pittsburgh

u/_agrippa_ · 13d

Have any of yinz non-native Pittsburgher notice any weird quirks that Pittsburghers do?

I had a coworker that moved here from the So.West. When I asked him this question he said "Pittsburghers had a habit of when after being asked a question, they would just stop and stare off into the distance for a good ten seconds before saying anything." I had to laugh because people in family do this. Me and my coworker now refer to this as the "Pittsburgh pause".

👍 812

👎

💬 842

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hegetsus.com

Learn More

 Jamiroquais_Dune · 13d

I've noticed almost everyone here has a lifelord nemesis in their neighborhood

⋮ ↩ Reply 🤖 👍 376 👎

Join the conversation

←

 r/pittsburgh · 20 hr. ago

blorfie

⋮

Yinz turning your heat back on?

I know, it takes two seconds, but it's the principle of the thing. It's almost June. I've had the A/C on a couple times already. It just doesn't feel right.

Like, don't get me wrong, I'm about to do it to maintain my marital harmony, but I can't think of another year where it's been this chilly this late in the season. Anyone toughing it out?

👍 258

👎

💬 228

🤖

🔗 Share

Join the conversation

Sort by: Best

🔍 Search Comments

 dfiler · 19h ago

It takes very little energy to heat your home a few more degrees when its only slightly cold outside. So I treat myself to the luxury of heat. It costs so much less to be comfortable on days like this than in the middle of winter.

⋮ 👍 155 👎 Reply 🤖 Award 🔗 Share ⋮

 manatee-manatou · 15h ago

Same here. I'll pay the few bucks it costs to take my tiny little cold-plaster-walled house from 60 degrees to 67 degrees so I'm not physically miserable all day. I'm far too miserable in other areas of my life to not turn the heat back on on days like today 😊

⋮ 👍 41 👎 Reply 🤖 Award 🔗 Share ⋮

 Junior_Bill_4691 · 15h ago

The correct response 👍

👍 8 👎 Reply 🤖 Award 🔗 Share ⋮

 ekpheartbooks · 15h ago

How is 'Yinzer' performed /r/Pittsburgh?

How many threads contain Pittsburghese language characteristics?

How does performing 'Yinzer' impact engagement?

What themes and contents are in these threads?

What sentiments and affects appear?

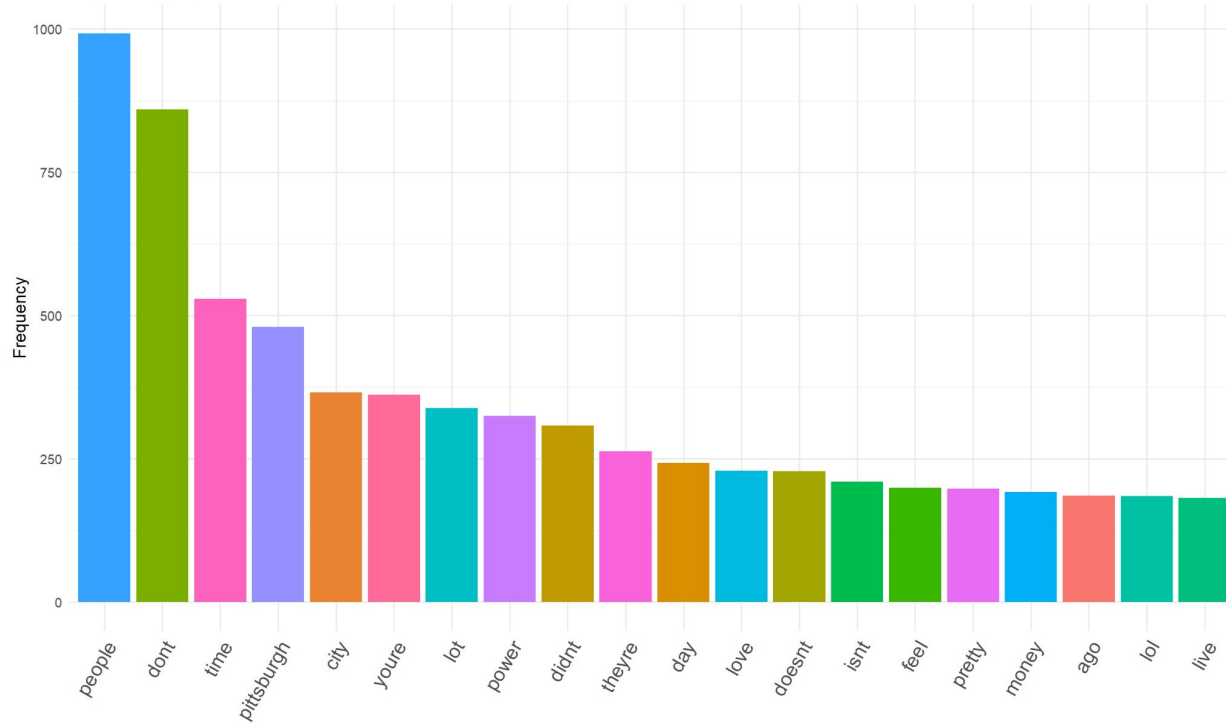
What do the sentiments identified in our analysis mean?

What relationships exist between sentiments and topics?



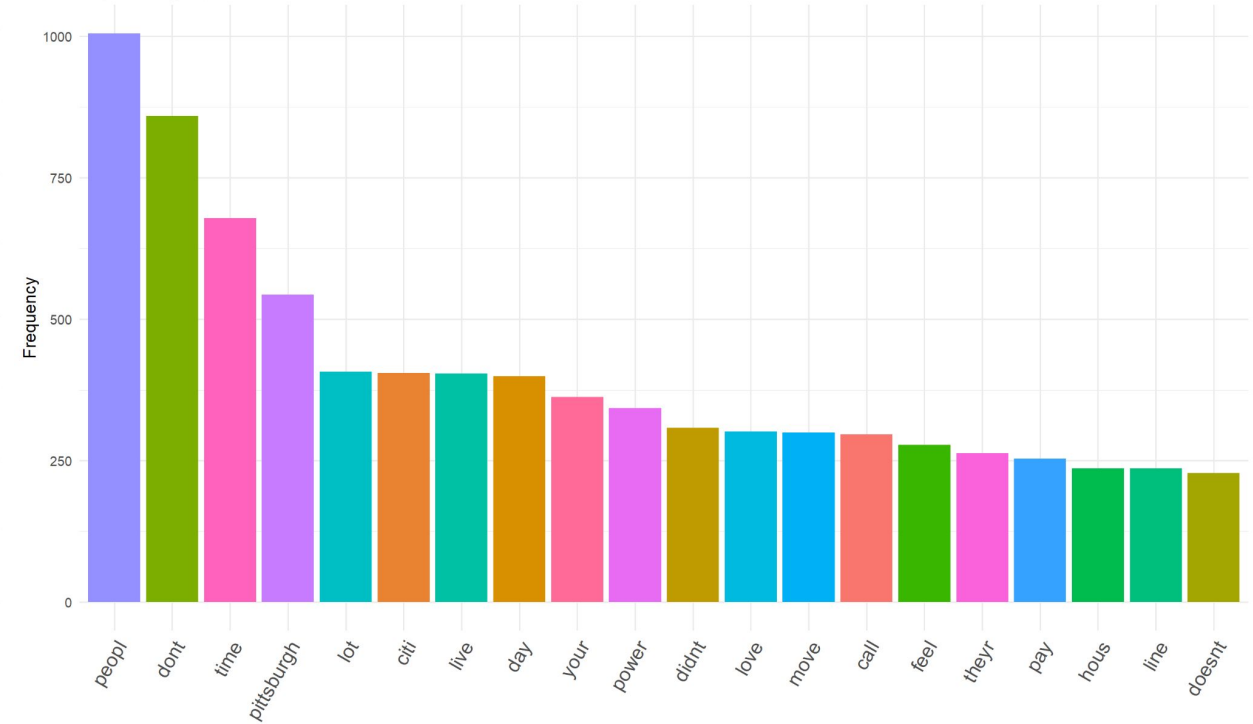
20 Most Frequent Words in Pittsburgh SubReddit Comments

May 4 to May 20, 2025

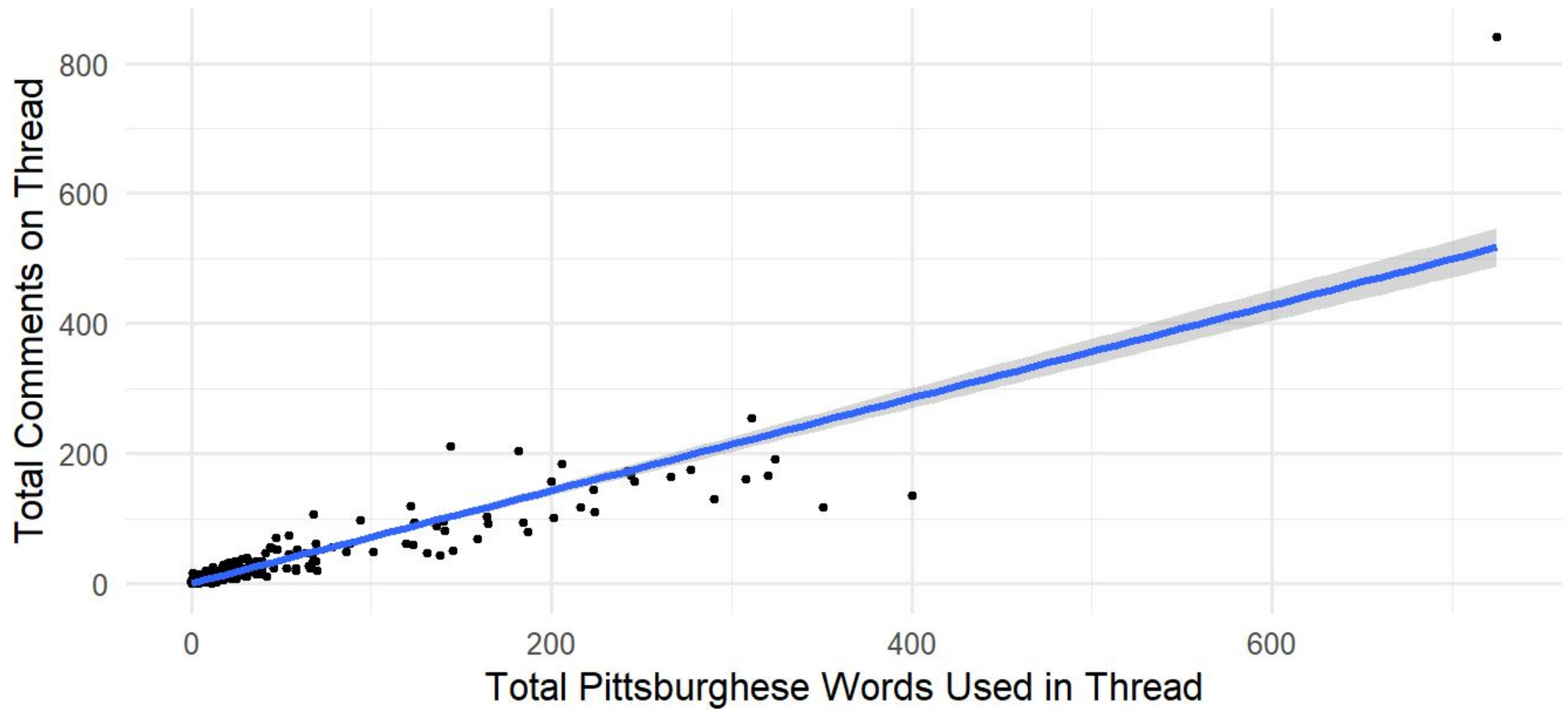


20 Most Frequent Stemmed Words in Pittsburgh SubReddit Comments

May 4 to May 20, 2025



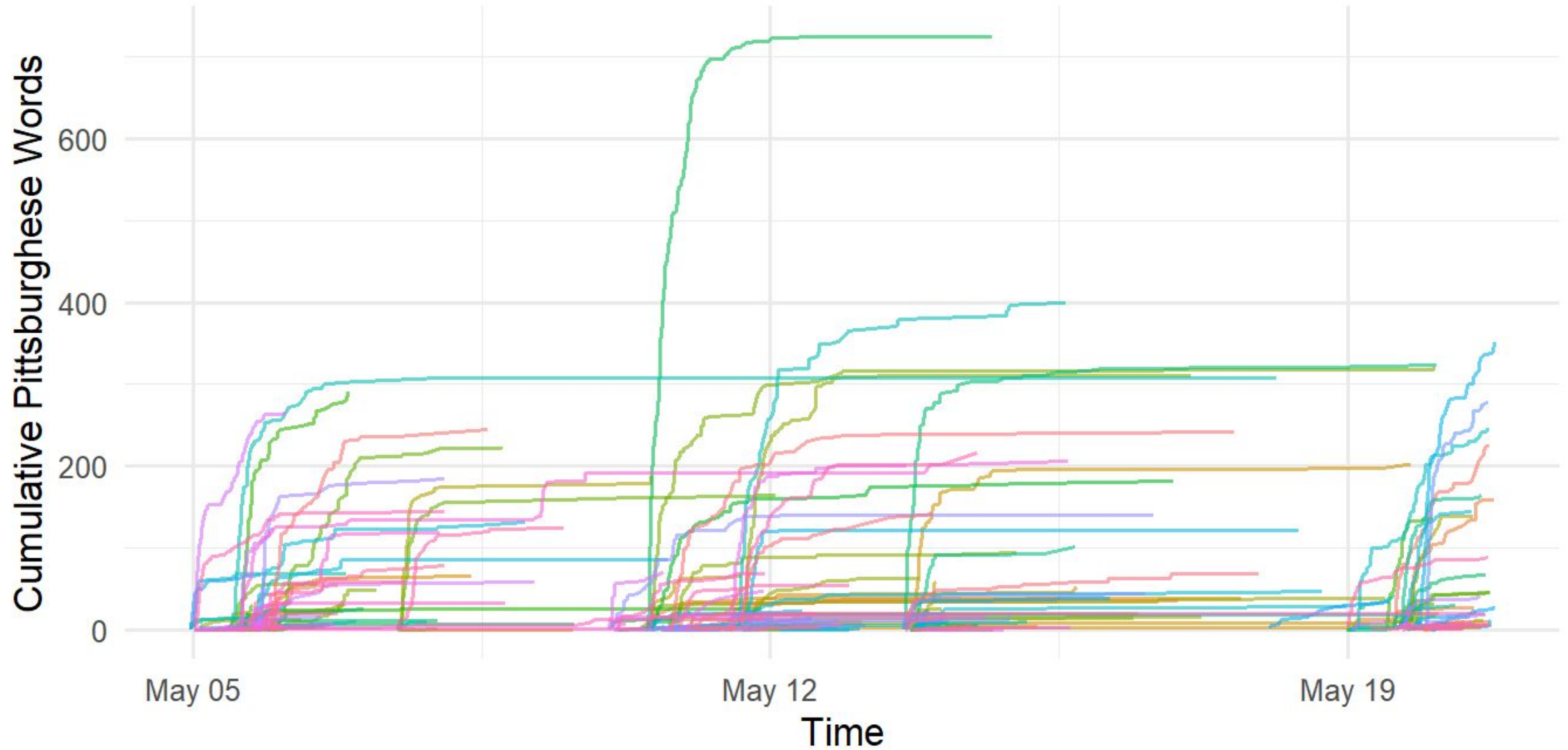
Thread Comments Vs Total Count of Pittsburghese Words



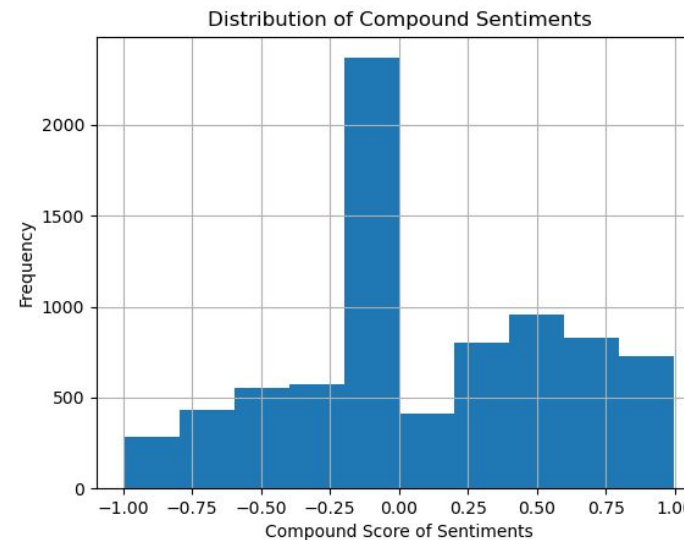
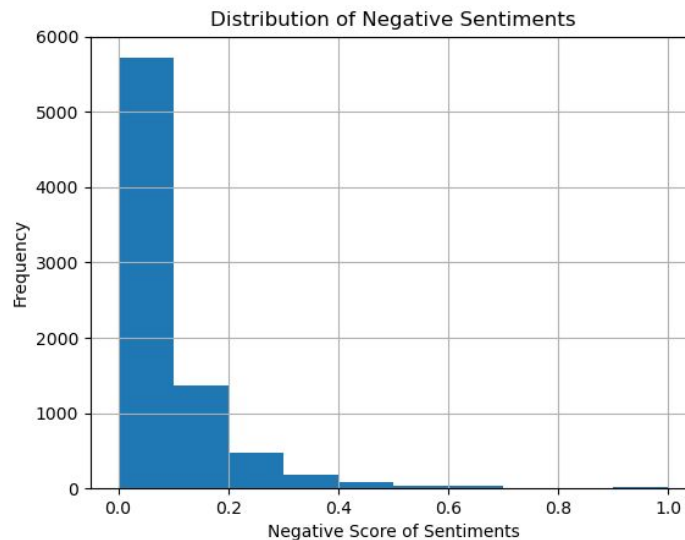
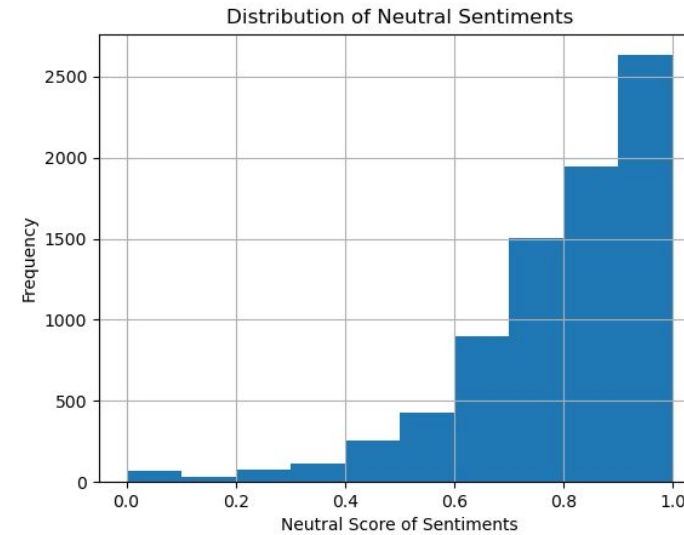
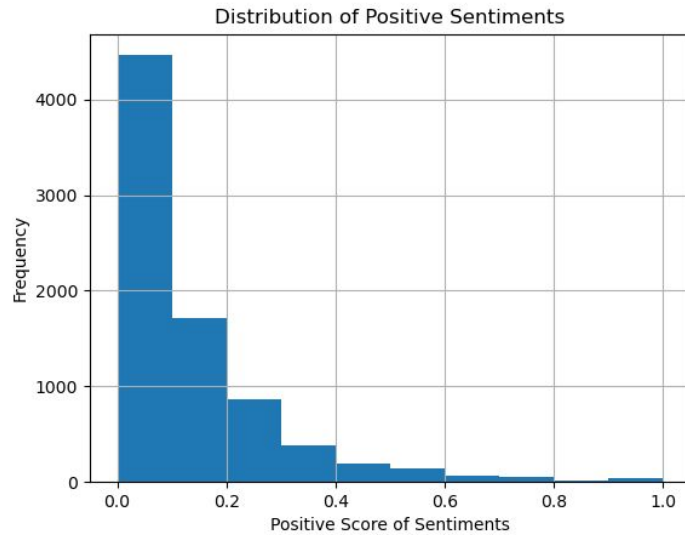
n threads = 266



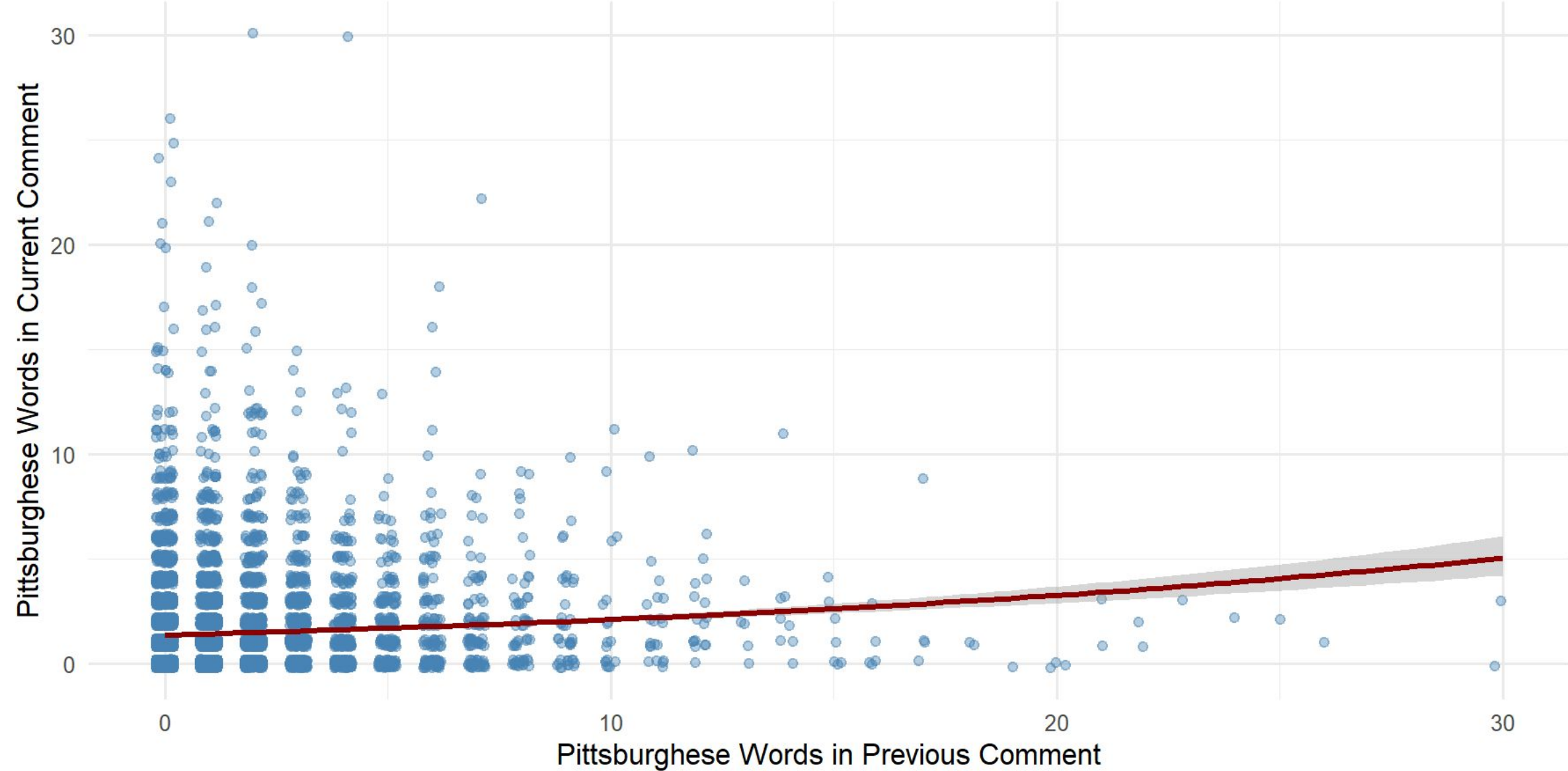
Cumulative Use of Pittsburghese Over Time



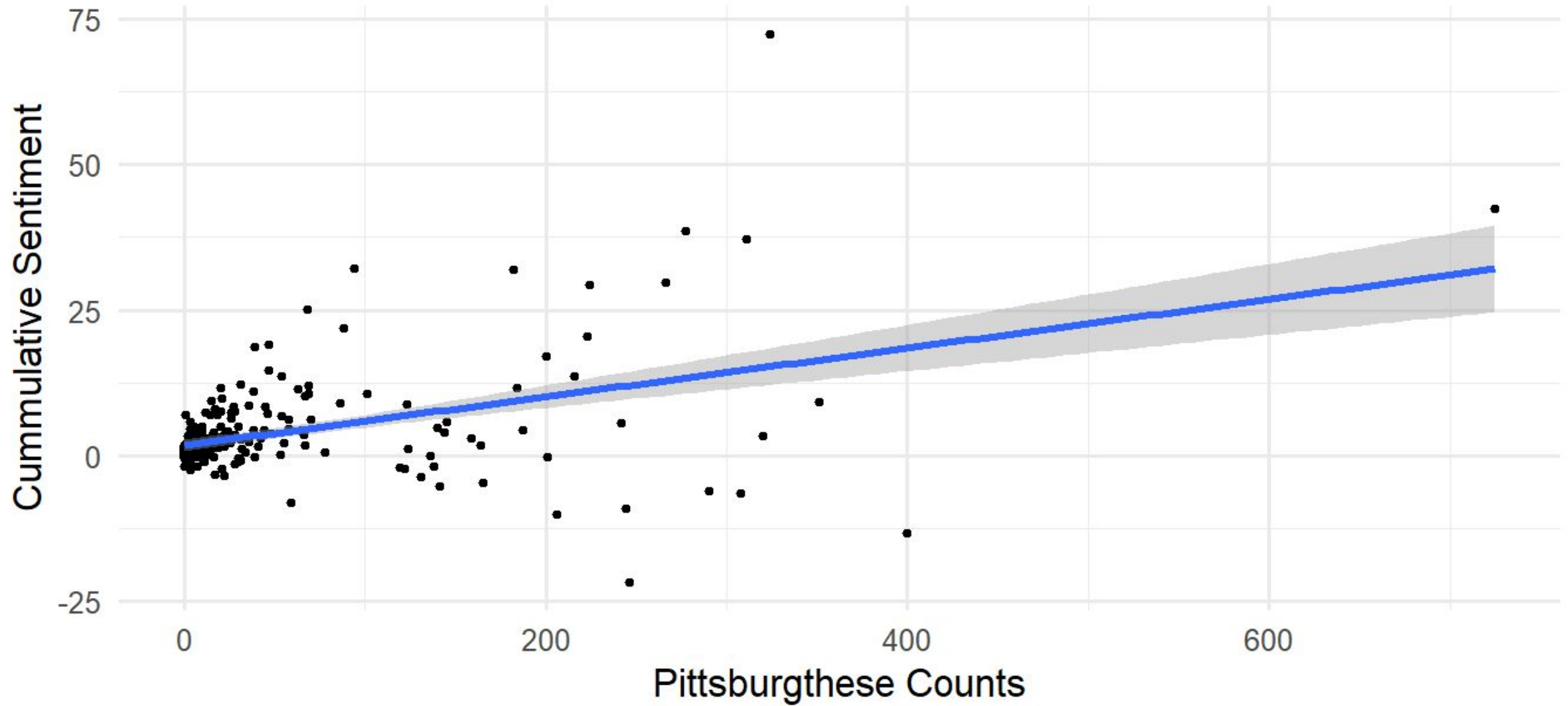
Sentiment Analysis



Correlation between Pittsburghese Count in Previous and Current Comment



Thread Comments Vs Total Count of Pittsburghese Words



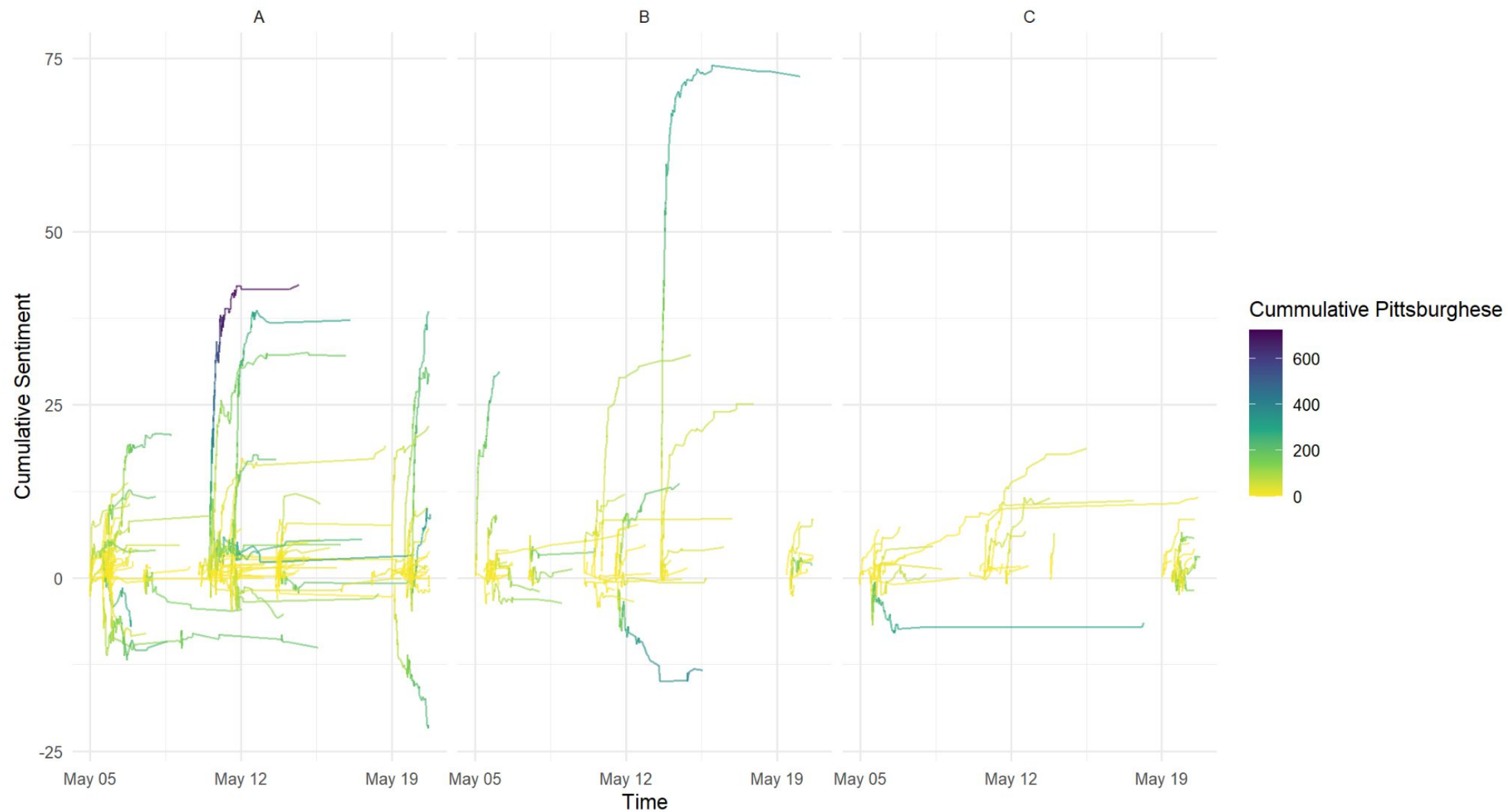
n threads = 266



Cumulative Sentiment and Pittsburghese Word Count Over Time



Cumulative Sentiment and Pittsburghese Word Count Over Time by Topic



Topic Modeling: K=3

```
# Compare how many unique words among 3 topics  
length(setdiff(all_terms_k3[, 1], all_terms[, 2])) # 712  
  
length(setdiff(all_terms_k3[, 2], all_terms[, 3])) # 703  
  
length(setdiff(all_terms_k3[, 1], all_terms[, 3])) # 712
```

All Terms Topic Modeling K3				
		Topic 1	Topic 2	Topic 3
1	1	pittsburgh	say	like
2	2	thing	people	people
3	3	just	pittsburgh	get
4	4	like	like	ive
5	5	needs	still	someone
6	6	one	just	pittsburgh

imagine that you are a creative anthropologists and you need to come up with a topic name based on a bundle of words for each topic column. please provide 5 possible topic/theme ideas for each column:

Topic 1:

Words include: pittsburgh, thing, just, like, needs, say, people, moved, never, know

Themes:

1. "Becoming Pittsburgh: Narratives of Arrival and Belonging"
2. "Saying What Matters: Everyday Language, Everyday Needs"
3. "Almost Home: The Emotional Cartography of Urban Settlement"
4. "First Words in a New Place: Speech, Memory, and Migration"
5. "Known Unknowns: Unspoken Needs in the Making of Place"

Topic 2:

Words include: say, people, pittsburgh, still, years, parents, moved, around, ive, always

Themes:

1. "Generations in Place: Family Histories and Urban Continuity"
2. "Echoes of Belonging: What We Still Say About Home"
3. "Temporal Anchors: Moving, Staying, and Saying So"
4. "Intergenerational Pittsburgh: Parental Presence and Place Memory"
5. "The Long Stay: Repetition, Place, and the Persistence of Self"

Topic 3:

Words include: like, people, get, someone, will, right, never, way, youre, know

Themes:

1. "Trying to Get It Right: Negotiating Belonging in Urban Talk"
2. "Somebody, Somewhere: Seeking Recognition in the City"
3. "Relational Maps: Knowing, Getting, and Being With Others"
4. "Vernacular Agency: What People Will and Won't Do"
5. "Making It Make Sense: The Struggle to Belong in Speech"

Limitations

Subject matter expertise: none of us are trained as sociolinguists

Data: Very small sample. Form of data (text) shapes questions and investigations. Inability to analyze audiovisual content in Reddit threads with methods we used

Demographics: This project does not account for Pittsburgh demographic profile, and only accounts for Reddit users

Technical skills: Mixed programming skills of team; some work done in Python

Text vs orality: Limits of capturing local language patterns solely through writing



Future Directions

Clean-up: Translate Python sections of our project into R

Data: Increase sample size (time duration)

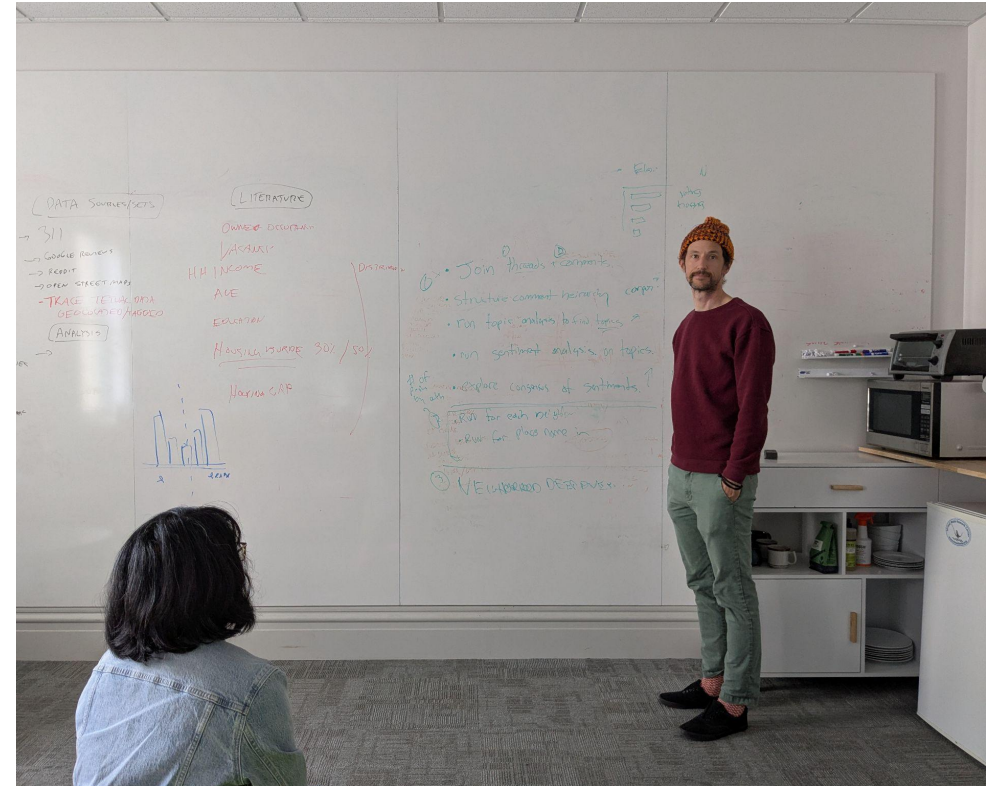
Comparative analysis: Use of Pittsburghese language referencing places (neighborhoods, geographic point)
Add map layer to visualizations

Network analysis: entities between user accounts

Time series: Examine user use patterns over time

Multimedia: Extract and analyze audiovisual media from Reddit posts

Subject matter: Read literature and learn more about Pittsburghese and sociolinguistics



Behind
the
scenes!



Literature

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Appendix

	thread_neg	thread_pos	thread_comp	thread_mean_sentiment	thread_pghese_count	thread_comments_num
thread_neg	1.0000000	0.79634720	0.2200455	-0.43785676	0.8187613	0.8894532
thread_pos	0.7963472	1.00000000	0.5663443	0.00387263	0.8053476	0.9171271
thread_comp	0.2200455	0.56634428	1.0000000	0.60967846	0.3999833	0.4435232
thread_mean_sentiment	-0.4378568	0.00387263	0.6096785	1.00000000	-0.1844709	-0.2260623
thread_pghese_count	0.8187613	0.80534762	0.3999833	-0.18447089	1.0000000	0.9082458
thread_comments_num	0.8894532	0.91712713	0.4435232	-0.22606235	0.9082458	1.0000000

```
> glimpse(comment_text)
Rows: 7,991
Columns: 9
$ datetime      <dtm> 2025-05-05 16:55:48, 2025-05-05 16:59:00, 2025-05-05 17:00:20, 2025-05-05 17:03:16, 2025-05-05 1...
$ text          <chr> "best of luck rehoming him and it was nice of you to take the stray in please try to really scree...
$ score         <dbl> 11, -14, 31, -33, 0, 31, 12, 2, 7, 2, 5, 1, 7, 1, 5, 11, 1, -18, 2, 1, 10, -13, 14, 6, 2, 4, 1, -...
$ thread_title  <chr> "1 month old male Kitten needs a home", "1 month old male Kitten needs a home", "1 month old male...
$ thread_score  <dbl> 65, 65, 65, 65, 65, 65, 65, 65, 65, 65, 65, 65, 65, 65, 65, 65, 65, 65, 65, 65, 65, 65, 65, 65, 6...
$ thread_comments_num <dbl> 45, 45, 45, 45, 45, 45, 45, 45, 45, 45, 45, 45, 45, 45, 45, 45, 45, 45, 45, 45, 45, 45, 45, 45, 4...
$ pghese_count  <int> 2, 2, 1, 0, 0, 1, 2, 1, 2, 1, 0, 0, 2, 0, 3, 2, 5, 6, 1, 1, 0, 1, 3, 0, 1, 1, 0, 2, 0, 2, 1, 0, 0...
$ lag_pghese_count <int> 0, 2, 2, 1, 0, 0, 1, 2, 1, 2, 1, 0, 0, 2, 0, 3, 2, 5, 6, 1, 1, 0, 1, 3, 0, 1, 1, 0, 2, 0, 2, 1, 0...
$ cum_pghese_count <int> 0, 2, 4, 5, 5, 6, 8, 9, 11, 12, 12, 12, 14, 14, 17, 19, 24, 30, 31, 32, 32, 33, 36, 36, 37, 38...
> pghese_inertia_model <- glm(pghese_count ~ lag_pghese_count + cum_pghese_count,
+                             data = comment_text,
+                             family = "poisson")
> summary(pghese_inertia_model)

Call:
glm(formula = pghese_count ~ lag_pghese_count + cum_pghese_count,
    family = "poisson", data = comment_text)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)   2.555e-01  1.275e-02  20.042   <2e-16 ***
lag_pghese_count 4.095e-02  3.336e-03  12.273   <2e-16 ***
cum_pghese_count 6.676e-04  7.122e-05   9.374   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 20494  on 7990  degrees of freedom
Residual deviance: 20262  on 7988  degrees of freedom
AIC: 32191

Number of Fisher Scoring iterations: 6
```



Appendix

```
> pghese_inertia_nb_model <- MASS::glm.nb(pghese_count ~ lag_pghese_count + cum_pghese_count,  
+                                         data = comment_text)  
> summary(pghese_inertia_nb_model)
```

Call:

```
MASS::glm.nb(formula = pghese_count ~ lag_pghese_count + cum_pghese_count,  
             data = comment_text, init.theta = 0.7925228755, link = log)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	0.2349785	0.0215812	10.888	< 2e-16	***
lag_pghese_count	0.0473688	0.0065250	7.260	3.88e-13	***
cum_pghese_count	0.0007778	0.0001286	6.049	1.46e-09	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for Negative Binomial(0.7925) family taken to be 1)

Null deviance: 8122.1 on 7990 degrees of freedom
Residual deviance: 8034.2 on 7988 degrees of freedom
AIC: 26453

Number of Fisher Scoring iterations: 1

Theta: 0.7925
Std. Err.: 0.0228

2 x log-likelihood: -26445.4340



github.com/ivan-rivera/RedditExtractor

README Code of conduct GPL-3.0 license

R-CMD-check no status

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Summary

Reddit Extractor is an R package for extracting data out of Reddit. It allows you to:

1. find subreddits based on a search query
2. find a user and their Reddit history
3. find URLs to threads of interest and retrieve comments out of these threads

Installation

The package can be installed directly from CRAN, using `install.packages("RedditExtractor")` or directly from GitHub via `devtools::install_github('ivan-rivera/RedditExtractor')`. Note that the latest version of this package requires R 4.1. If you have an earlier version and are not ready to upgrade, then you can install the earlier version of this package with `devtools::install_version("RedditExtractor", version = "2.1.5", repos = "http://cran.us.r-project.org")`. Beware that version 3+ introduces significant breaking changes!

Quick start

Let's suppose that I'd like to get the top posts from the [r/cats](#) subreddit. After importing the package with `library(RedditExtractor)` here is how we can do it:

Languages
R 100.0%



Thank You! :-)

